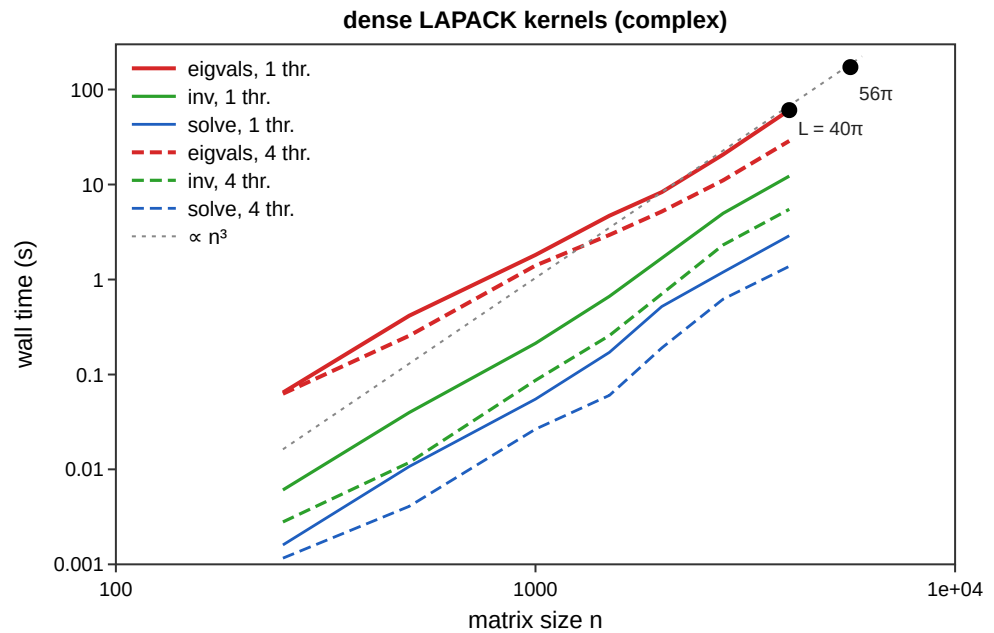
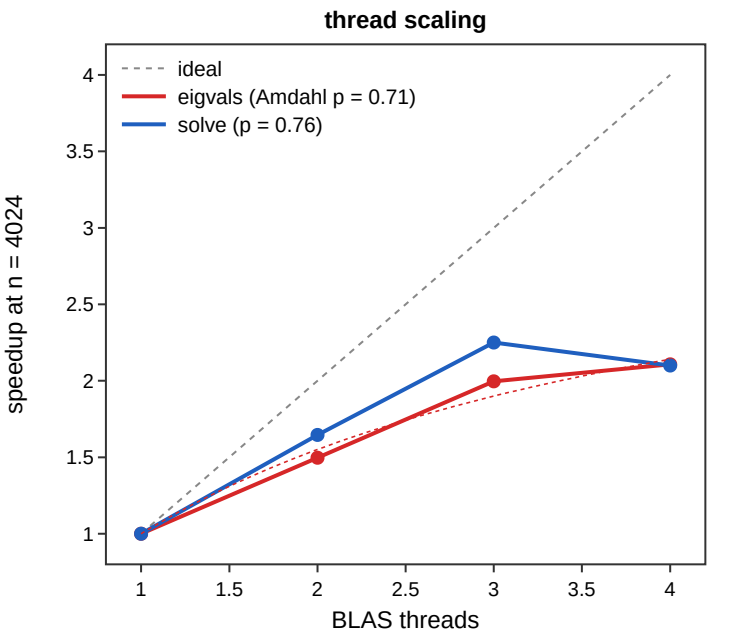

dense LAPACK kernels (complex)
eigvals, 1 thr.
inv, 1 thr.
solve, 1 thr.
eigvals, 4 thr.
inv, 4 thr.
solve, 4 thr.
∝ n³
56π
L = 40π
wall time (s)
100
10
1
0.1
0.01
0.001
100
1000
1e+04
matrix size n


thread scaling
ideal
eigvals (Amdahl p = 0.71)
solve (p = 0.76)
speedup at n = 4024
4
3.5
3
2.5
2
1.5
1
1
1.5
2
2.5
3
3.5
4
BLAS threads